Andrei-Cristian Cojocaru

Berkeley, CA | cojoandrei03@berkeley.edu | Website | GitHub | Google Scholar | LinkedIn

Education

University of California, Berkeley *Expected May 2027*

B.A. in Computer Science

Coursework: NLP, Reinforcement Learning, Deep Learning, Distributed Systems, Information Theory, Convex Optimization

Research Interests

Machine learning and AI systems, with research in evaluation, optimization, and learning from structured feedback. Broader interests include reinforcement learning, continual learning, simulation, and world models.

Research Experience

BAIR and Sky Computing Lab, UC Berkeley *April 2025 - Present*

Undergraduate Researcher

AdaMAST *November 2025 - Present*

Equal-contribution co-first author | Advisor: Ion Stoica | Code

- Developed the adaptive-taxonomy approach with Mert Cemri from his initial concept. Took primary responsibility for methods that derive failure categories from execution evidence and refine them as programs evolve.
- Built the end-to-end taxonomy pipeline and its agreement-based and reflection-based refinement methods, including category generation, validation, merging, splitting, and revision from execution traces and annotations.
- Developed failure-labeling and causal-analysis judges with category-coverage and annotation-quality checks. Built AdaEvolve and live-runtime integrations to supply diagnoses during candidate optimization and agent execution.
- Independently proposed and built Claude Code/Codex checkpoints, a submission gate, and in-session taxonomy learning; released them in the adamast PyPI package. Contributed to paper formulation and writing.

GEPA Research and Optimization *June 2026 - Present*

Equal-contribution first author of research article | With the GEPA team | Faculty collaborator: Matei Zaharia | Code

- Proposed and built learned error diagnosis for GEPA's reflection, including taxonomy generation and reduction, component-specific diagnosis, trace caching, parallel judging, cost accounting, and experimental evaluation.
- Designed controlled, held-out comparisons with baseline GEPA under matched online budgets. Kept the diagnosis outcome-blind while retaining gold-based evaluation and selection in the published reflection study.
- Integrated the method with GEPA adapters and documented its experimental workflow in the research article and open-source release. Ongoing extensions investigate gold-free evaluation and optimization.

ProofGrader and ProofBench *April - September 2025*

Co-author | Advisor: Sewon Min | With Wenjie Ma and Matei Zaharia

- Contributed to ideation with Sewon Min and Wenjie Ma and built the first marking-scheme generator. Refined criteria for partial credit and valid alternative approaches through manual evaluation of Olympiad-level proofs.
- Produced expert annotations and qualitative analyses for ProofBench, distinguishing mathematical reasoning errors from limitations in grading criteria and mistakes in the LLM judge's interpretation.
- Developed and evaluated ProofGrader variants in parallel with Wenjie Ma, combining changes after joint analysis. Contributed to final ablations and manuscript revisions, and presented the work as a poster at ICLR 2026.

Ongoing Research

SFT Relay with AdaMAST *In progress (unpublished)*

- Developing supervised fine-tuning for coding agents with failure signals that guide teacher intervention.

FailureRank and Related Research *August 2026 - Present*

Lead researcher on FailureRank | Advisor: Ion Stoica

- Developing FailureRank to rank LLMs and agents from judge-identified failure modes using a handful of graded tasks. Learning failure-mode weights without outcome labels using item response theory with a bifactor habit structure and trace length as a covariate.
- Validating rankings in simulation and on agentic benchmarks against held-out pass rates, especially when small graded samples misrank candidates. Using prediction-powered inference to quantify uncertainty.
- Developing purpose-specific taxonomies and judges for correctness, robustness, and required behaviors. Exploring reward evaluation and structured feedback in RL, evolutionary optimization, and continual learning.

Publications and Research Articles

[1] **Reliable Fine-Grained Evaluation of Natural Language Math Proofs**

W. Ma, **A. Cojocaru**, N. Kolhe, B. Louie, R. S. Sharif, H. Zhang, V. Zhuang, M. Zaharia, and S. Min.

ICLR 2026. Conference paper. Paper

[2] **Fantastic Adaptive Taxonomies and How to Use Them**

M. Cemri*, **A. Cojocaru***, M. Pan, S. Liu, S. Agarwal, A. Krentsel, J. Tang, K. Ramchandran, J. E. Gonzalez, M. Zaharia, A. Dimakis, and I. Stoica.

FAGEN Workshop at ICML 2026. **Best Paper Award.** Paper

[3] **Learned Error Diagnosis Improves GEPA's Reflection**

A. Cojocaru*, L. A. Agrawal*, M. Cemri, J. E. Gonzalez, A. G. Dimakis, I. Stoica, and M. Zaharia.

GEPA Blog, October 2026. Research article and reproducibility release. Article

* Equal contribution. ICLR 2026: presented the ProofGrader paper as a poster.

Industry Experience

Mercor | *Machine Learning Developer* *May - November 2025*

- Analyzed model traces and candidate outputs, manually produced correct reference responses, and supplied feedback to improve evaluation quality and model behavior.

Logitech | *Software Engineering Intern* *July - December 2024*

- Developed an AI-assisted cloud-resource optimization system using AWS EC2, Cost Explorer, SageMaker, and Lambda; analyzed operational data with SQL and Spark to inform resource-allocation decisions.

Outlier AI | *Machine Learning Developer* *April - November 2024*

- Analyzed model behavior and provided evaluation feedback for ML research and iterative improvement.

Makers Fund | *Contracted Research Project* *May - July 2024*

- Built a market-intelligence pipeline for blockchain gaming using DappRadar, Artemis, CoinGecko, spaCy, and Pandas to produce research summaries and market signals for investment analysis.

Technical Skills

ML and NLP: Model training, supervised fine-tuning, LLM post-training, reinforcement learning, sentiment analysis, summarization

Programming languages: Python, C++, SQL, TypeScript, Java

ML libraries: PyTorch, TensorFlow, Hugging Face Transformers and Datasets, DSPy, LiteLLM, scikit-learn, spaCy, OpenCV

Data and cloud: NumPy, Pandas, Matplotlib, Spark, MongoDB, Airflow; AWS (EC2, SageMaker, Lambda, Bedrock, Cost Explorer)

Development tools: Git, GitHub Actions (CI/CD), Docker, Linux, pytest, Ruff, uv, Django, Node.js